

JKBOSE 2026

CLASS 11TH

PLANT KINGDOM

LECTURE-01

BOTANY

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Topics To Covered

- 1** Introduction ✓
- 2** Plant Classification ✓
- 3** Branches of Taxonomy ✓

INTRODUCTION

≡ Plant Kingdom — includes all those organisms which are **Eukaryotic** and **chlorophyll** bearing.

≡ Cell wall is **thick** — mainly composed of **cellulose**.

≡ Plants are **Autotrophic**. Some plants are **partially heterotrophic**.
↓ like

a. insectivorous plants ex. Venus fly trap
Bladderwort

b. Parasitic plants ex. Cuscuta.

PLANT CLASSIFICATION

1. Artificial classification.
2. Natural classification.
3. Phylogenetic classification.

(1) ARTIFICIAL CLASSIFICATION

≡ Earliest System of Classification.

≡ Here plants are classified on the basis of Gross morphological Characters like habit, no. and shape of leaves etc.

• Example:

① Theoprastus (father of Botany)

↓
Classified plants into 04 groups
on the basis of habit.

① Herbs

② Under Shrubs

③ Shrubs

④ Trees

② C. Linnaeus (father of taxonomy).

↓
Classified flowering plants (Angiosperms) into 24 Classes on the basis of no. of Stamen.

* Drawbacks of Artificial Classification :-

①. No use of Phyllogeny (i.e. evolutionary relationship among organisms).

②. Equal weightage given to reproductive and vegetative parts
(vegetative characters get easily affected by environmental conditions).

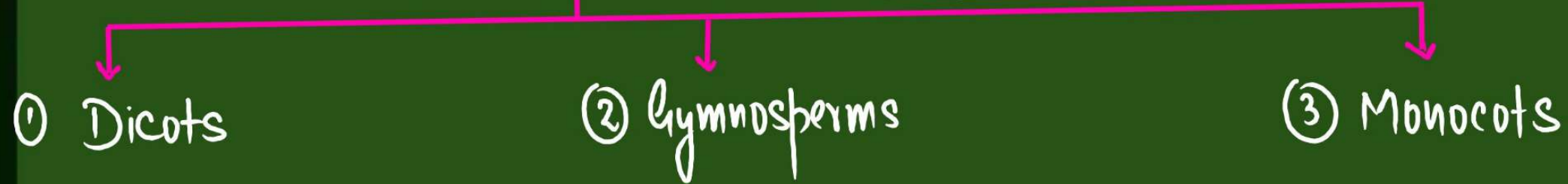
(2) NATURAL CLASSIFICATION

• Here plants are classified on the basis of :

- a. Morphological characters.
- b. Anatomical characters.
- c. Cytological characters
- d. Embryological Characters
- e. Physiological characters etc. etc.

Example: C. Bentham and J.D Hooker

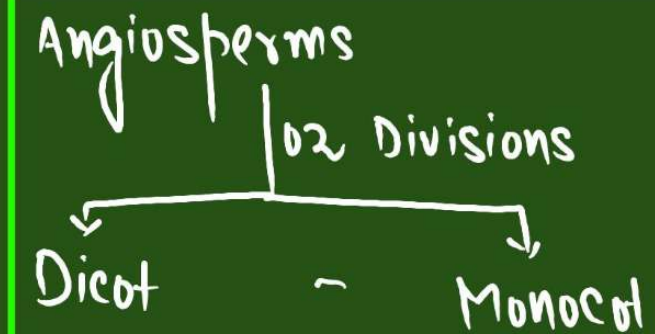
classified Spermatophytes (seed bearing plants)
into 03 groups.



Drawbacks of Natural classification:

- ① Placement of Gymnosperms b/w monocots and Dicots was not acceptable
- ② No use of phylogeny.

Remember*



QUESTIONS

Classification given by Bentham and Hooker is:

[1988]

- 1 Artificial
- 2 Natural
- 3 Phylogenetic
- 4 Numerical

(3) PHYLLOGENETIC CLASSIFICATION

- It is AKA Cladistics.
 - Here plants are classified on the basis of **Evolutionary relationship** along with – morphological characters
Anatomical characters
Cytological characters etc etc.
- ∴ Most acceptable classification.

Plant Kingdom

Phylogenetic Classification

1. Cryptogamae

- Seed less plants.
- Sex organs - hidden

Thallophyta (Algae)

Bryophyta

Pteridophyta

2. Phanerogamae

- Seed bearing plants
- Sex organs - well exposed

Gymnosperm

< Gymno's = Naked
Sperma = Seed

Angiosperm

- Flowering plants
- < Angios = covered
Sperma = Seed

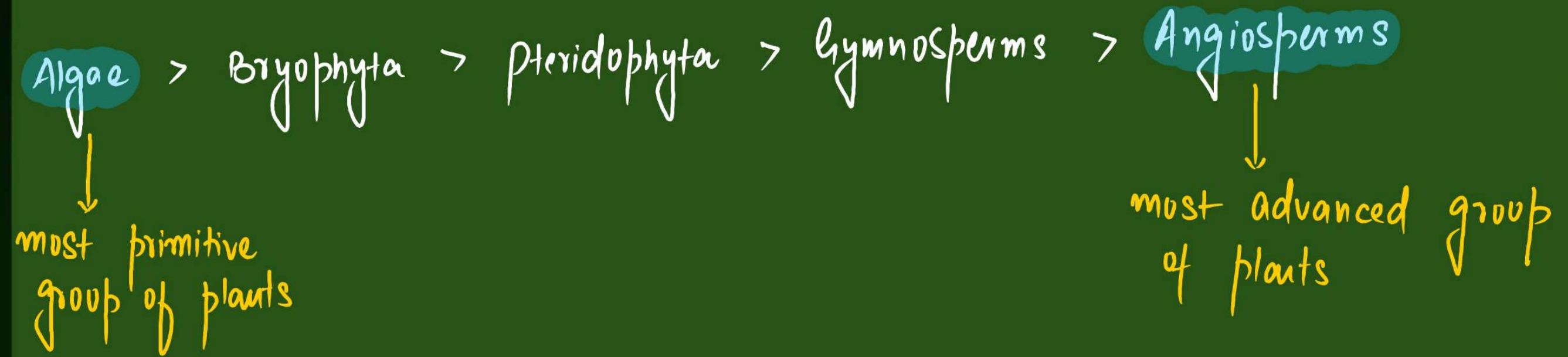
Dicots

(Seed having 02 cotyledons)

Monocots

(Seed having 01 cotyledon)

AS per Evolution:



BRANCHES OF TAXONOMY

• 03 Branches :

- ① Chemotaxonomy.
- ② Cytotaxonomy.
- ③ Numerical taxonomy.

Taxonomy



Branch of Science
which deals with
classification of living
organisms.

Chemotaxonomy

- plants classified on the basis of chemical affinities.
- Chemicals include:
Resins,
tanins,
phenols,
alkaloids etc

Cytotaxonomy

• plants classified on the basis of Cytological characters like:

- Chromosome no,
- behaviour of chromosomes during cell division.

Numerical Taxonomy

- plants are classified on the basis of all the observable characters.
- Each character is given equal weightage.
- No. & nodes are assigned to each character.
- The data is then processed by using computers.

QUESTIONS

System of classification used by Linnaeus are:

[1989]

- 1 Natural system
- ✓ 2 Artificial system
- 3 Asexual system
- 4 Phylogenetic system

QUESTIONS

Artificial system of classification was first used by:

[1989]

- ☒ 1 Linnaeus
- ☐ 2 de Candolle
- ☐ 3 Pliny the Elder
- ☐ 4 Bentham and Hooker

QUESTIONS

Phylogenetic system of classification is based on:

[2009]

- 1 Morphological features
- 2 Chemical constituents
- 3 Evolutionary relationships
- 4 Floral characters

Thank
You